

Hein Zarni Aung

heinzarniaung.dev@gmail.com | linkedin.com/in/heaung | github.com/LoistCloist

EDUCATION

Stony Brook University

Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Algorithms, Programming Abstractions

INTERNSHIPS

Software Engineering Intern — Clustr

Jan 2026 - Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.
- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.

Software Engineering Intern — Prenora Dynamics LLC

Jun 2026 - Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, and built a seed-data workflow to streamline debugging and local development.

PROJECTS

TuneAcademy

React, NumPy, Essentia, Firestore, FastAPI

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy the app to Cloudflare Workers.

Discord Voice Channel Activity Monitor

Node.js, Discord.js, PM2, AWS

- Identified a recurring coordination problem for a friend group spread across different schedules and built a Node.js/Discord.js bot to automate real-time notifications when someone joins a voice channel, removing the need to manually check in.
- Deployed on a cloud server using PM2; reached 50 concurrent active users.

CAMPUS INVOLVEMENT

Teaching Assistant - Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University

Aug 2025 - Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium